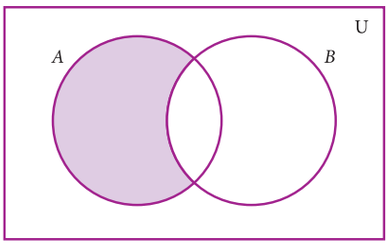
Example 1.16

Represent $A\Delta B$ through Venn diagram.

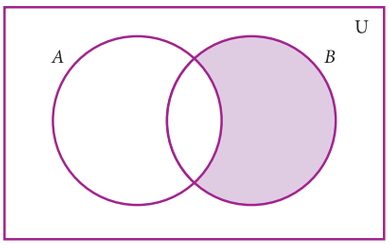
Solution

$A\Delta B = (A - B) \cup (B - A)$



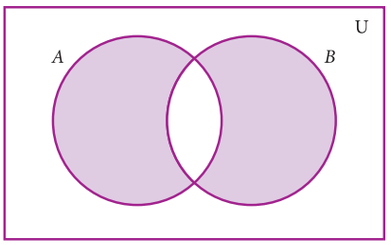
$A - B$

Fig. 1.16



$B - A$

Fig. 1.17



$(A - B) \cup (B - A)$

Fig. 1.18

Note

- $A\Delta A = \varnothing$
- $A\Delta B = B\Delta A$
- $A\Delta B = \{x : x \in A \cup B \text{ and } x \notin A \cap B\}$
- $A\Delta B = (A \cup B) - (A \cap B)$

Example 1.17

From the given Venn diagram, write the elements of

(i) A (ii) B (iii) $A - B$ (iv) $B - A$

(v) A' (vi) B' (vii) U

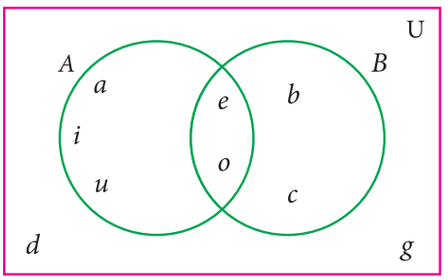


Fig. 1.19

Solution

(i) $A = \{a, e, i, o, u\}$

(ii) $B = \{b, c, e, o\}$

(iii) $A - B = \{a, i, u\}$

(iv) $B - A = \{b, c\}$

(v) $A' = \{b, c, d, g\}$

(vi) $B' = \{a, d, g, i, u\}$

(vii) $U = \{a, b, c, d, e, g, i, o, u\}$

Example 1.18

Draw Venn diagram and shade the region representing the following sets

(i) A' (ii) $(A - B)'$ (iii) $(A \cup B)'$

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